

A Conditional Construction of Lorentzian Geometry

from Euclidean Three-Space: Clifford Double-Cover Data, Gluing, and a Soldering Isomorphism

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Abstract

We study a conditional construction whose local input is only the standard positive-definite Euclidean space $V = \mathbb{R}^3$. From this datum one constructs the real spin factor $\mathcal{S} = \mathbb{R} \oplus V$, its quadratic form $N(a, v) = a^2 - \langle v, v \rangle$ of signature $(1, 3)$, the real Clifford algebra $\text{Cl}(V, q)$, a Clifford norm-one group G_{Spin} , and a surjective central two-fold covering

$$1 \longrightarrow \{\pm 1\} \longrightarrow G_{\text{Spin}} \xrightarrow{\rho} G_{\text{Lor}} \longrightarrow 1$$

of the proper orthochronous Lorentz group $G_{\text{Lor}} \subset O(B)$ defined below. The B -skew-adjoint endomorphisms of $\mathcal{S}$ form the six-dimensional orthogonal Lie algebra $\mathfrak{so}(B)$, canonically isomorphic to $\Lambda^2 \mathcal{S}$.

These local constructions do not determine a global manifold. We therefore introduce explicit gluing data. If the graph of the gluing equivalence relation is closed, the index set is countable, and the transition maps are smooth, the quotient is a Hausdorff second-countable smooth four-manifold M .

Independently, a G_{Spin} -valued Čech 1-cocycle projects to a Lorentz-valued cocycle. This cocycle need not coincide with the tangent transition cocycle $D\phi_{ij}$. Additional smooth soldering data intertwine the two cocycles and determine a vector-bundle isomorphism from the associated Lorentz vector bundle $E_g \rightarrow M$ to TM ; the inverse is the conventional E_g -valued solder form $TM \rightarrow E_g$. These data induce a smooth Lorentzian metric on M , orientation compatibility, and a continuous compatible choice of the future timelike cone component in each tangent fibre. They also imply that the determinants of the tangent transition maps form a multiplicative Čech 1-cocycle that is a continuous coboundary and that the $\{\pm 1\}$ -valued Čech 2-cocycle class associated with the central extension is trivial on the chosen cover. No identification of these classes on the chosen cover with $w_1(TM)$ or $w_2(TM)$ is made.

Finally, on an explicit periodic gluing we study a one-parameter family of G_{Spin} -valued Čech 1-cocycles. Soldering data exist for every finite parameter. Any two members are gauge equivalent under G_{Spin} -valued Čech 0-cochains; the corresponding sets of local soldering data are explicitly bijective; and corresponding choices of soldering data induce the same Lorentzian metric. This result is restricted to that family and does not construct a connection, holonomy, curvature, torsion, or dynamics.

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1 Introduction

Standard Lorentzian spin geometry is usually formulated after a smooth manifold, a Lorentz metric, and the associated frame bundle have already been supplied; the Spin structure then appears as a lift of an orthonormal frame bundle subject to a topological obstruction [6, 8, 11]. Here the order of dependence is deliberately reversed. The starting object is only

$$V = \mathbb{R}^3, \quad \langle u, v \rangle = u_1 v_1 + u_2 v_2 + u_3 v_3,$$

with no four-dimensional base space, tangent bundle, Lorentz form, orientation, Spin bundle, connection, or curvature in the input.

There are then three logically separate construction questions.

First, what local Lorentzian and Clifford-algebraic structures can be constructed from the local Euclidean algebra? Second, what information must be added before local copies of the resulting model can be assembled into a global smooth manifold? Third, after such a manifold has been obtained, what additional datum is required to identify the Lorentz-valued cocycle with the tangent transition cocycle?

The separation of these questions is essential. A local model is not a manifold; fibre transition data do not determine which local points are to be identified; and a Lorentz-valued cocycle is not, without further structure, the derivative cocycle of a manifold atlas. We make each of these boundaries explicit and place the positive construction theorems immediately after the relevant missing datum.

The local algebra belongs to the standard circle of ideas around real Clifford algebras, Spin groups, spin factors and Lorentz cones [1, 6, 4, 2]. The global steps use standard quotient, manifold, bundle and Čech ideas [7, 3, 11, 5], but the point of the present organization is not to assume the usual global Lorentzian structures in advance. Instead, the precise dependency chain is

$$\begin{array}{c}
 (\mathbb{R}^3, \langle \cdot, \cdot \rangle) \\
 \Downarrow \\
 (\mathcal{S}, N, B, \text{Cl}_3, G_{\text{Spin}}, \rho, \mathfrak{so}(B)) \\
 + \text{gluing data} \\
 \Downarrow \\
 \text{quotient covered by coordinate domains modelled on } \mathcal{S} \\
 + \text{closed graph of the gluing relation} + \text{countable index} + \text{smooth transition maps} \\
 \Downarrow \\
 M^4, \quad TM, \quad D\phi_{ij} \\
 + \text{soldering data} \\
 \Downarrow \\
 (TM, g), \\
 \text{orientation compatibility,} \\
 \text{compatible choice of a timelike cone component in each tangent fibre,} \\
 \text{tangent cocycle in the local frames defined by } A_i = \text{projected Lorentz cocycle}
 \end{array} \tag{1.1}$$

The terms preceded by plus signs are additional data or hypotheses; they are not derived from the preceding local algebra.

The following six statements organize the paper.

Theorem 1.1 (Local construction). *From the specified Euclidean datum $V = \mathbb{R}^3$, the construction produces:*

1. a four-dimensional real spin factor $\mathcal{S} = \mathbb{R} \oplus V$ whose polarized quadratic form has signature $(1, 3)$;

2. the real Clifford algebra $\text{Cl}_3 = \text{Cl}(V, q)$, with the paravector embedding of $\mathcal{S}$;
3. a Clifford norm-one group G_{Spin} , the proper orthochronous Lorentz group $G_{\text{Lor}} \subset O(B)$ determined by the chosen future cone, and a continuous surjective homomorphism $\rho : G_{\text{Spin}} \rightarrow G_{\text{Lor}}$ whose central kernel is exactly $\{\pm 1\}$, together with continuous local sections;
4. the six-dimensional orthogonal Lie algebra $\mathfrak{so}(B)$ and a canonical linear isomorphism $\Lambda^2 \mathcal{S} \simeq \mathfrak{so}(B)$.

Theorem 1.2 (Globalization under explicit gluing hypotheses). *The local model does not determine a global quotient space. Given explicit compatible gluing data B such that the graph of the gluing equivalence relation is closed, the index set is countable, and the transition maps are smooth, the quotient M_B is a Hausdorff second-countable smooth four-manifold modelled on $\mathcal{S}$.*

Theorem 1.3 (Independence of tangent and Lorentz-valued cocycles). *For the constructed smooth manifold, the tangent transition cocycle is $D\phi_{ij}$. Independently, a G_{Spin} -valued Čech 1-cocycle projects to a Lorentz-valued cocycle $\rho(\tilde{g}_{ij})$. Smooth gluing together with the latter cocycle does not force the two cocycles to agree.*

Theorem 1.4 (Soldering isomorphism). *Soldering data are an additional smooth local family of fibrewise linear isomorphisms satisfying the exact overlap intertwining law between $D\phi_{ij}$ and $\rho(\tilde{g}_{ij})$. Conditional on their existence, they yield a vector-bundle isomorphism between E_g , the Lorentz vector bundle defined by the cocycle, and TM , with continuous total-space maps in both directions and smooth local representatives. They also induce a smooth Lorentzian metric of signature $(1, 3)$.*

Theorem 1.5 (Intertwining of tangent and Lorentz cocycles). *For every choice of soldering data, the tangent transition cocycle expressed in the local frames defined by A_i equals the projected Lorentz-valued cocycle. Consequently the determinants of the tangent transition maps form a multiplicative Čech 1-cocycle that is a continuous coboundary, and the $\{\pm 1\}$ -valued Čech 2-cocycle class for the central extension is trivial on the chosen cover. The soldering data also yield a continuous compatible choice of the timelike cone component corresponding to C_{int}^+ in each tangent fibre.*

Theorem 1.6 (Gauge equivalence of the one-parameter family). *For the periodic gluing example and the one-parameter family of G_{Spin} -valued Čech 1-cocycles considered in section 11, every finite parameter value admits soldering data. Any two parameter values define cocycles gauge equivalent under G_{Spin} -valued Čech 0-cochains. The corresponding sets of local soldering data are explicitly bijective, and corresponding choices of soldering data induce the same Lorentzian metric.*

The hypotheses in theorems 1.2 and 1.4 are not cosmetic. They are exactly where the construction is no longer determined by the initial Euclidean local data. In particular, this paper does *not* prove an implication

$$\mathbb{R}^3 \implies \text{a uniquely determined smooth Lorentzian four-manifold.}$$

It proves the local implication in theorem 1.1, identifies the additional global data required to continue, and then proves what follows from those data.

2 Euclidean input, the spin factor, and its Lorentzian quadratic form

2.1 Euclidean input

Throughout the local part of the paper,

$$V := \mathbb{R}^3$$

carries only its standard positive-definite bilinear form

$$\langle u, v \rangle := u_1 v_1 + u_2 v_2 + u_3 v_3.$$

Set

$$q_3(v) := \langle v, v \rangle.$$

No Lorentzian structure has yet been introduced.

Definition 2.1 (Real spin factor). Define

$$\mathcal{S} := \mathbb{R} \oplus V$$

and, for $x = (a, u)$, $y = (b, v)$, define

$$x \circ y := (ab + \langle u, v \rangle, av + bu).$$

The distinguished unit is

$$\mathbf{1}_{\mathcal{S}} := (1, 0).$$

The product is bilinear and commutative. As an auxiliary local identity for this chosen spin-factor construction one also has the Jordan identity

$$(x \circ y) \circ (x \circ x) = x \circ (y \circ (x \circ x)).$$

Thus $(\mathcal{S}, \circ, \mathbf{1}_{\mathcal{S}})$ is the rank-two real spin factor; compare the standard Jordan-algebra treatment in [4, 2]. This local identity is supporting algebra, not an additional global construction claim.

Definition 2.2 (Quadratic and bilinear forms). For $x = (a, u) \in \mathcal{S}$, set

$$N(x) := a^2 - \langle u, u \rangle.$$

Its polarization is

$$B(x, y) := \frac{N(x + y) - N(x) - N(y)}{2} = ab - \langle u, v \rangle$$

for $y = (b, v)$.

Proposition 2.3 (Lorentz signature). *The form B is symmetric and nondegenerate, and has signature $(1, 3)$. In the standard basis*

$$\mathbf{1}_{\mathcal{S}}, \quad (0, \varepsilon_0), \quad (0, \varepsilon_1), \quad (0, \varepsilon_2),$$

its Gram matrix is

$$\text{diag}(1, -1, -1, -1).$$

Proof. The displayed formula $B((a, u), (b, v)) = ab - \langle u, v \rangle$ gives the Gram matrix directly. Since the Euclidean form on V is positive definite, the first basis direction is positive and the remaining three are negative. The determinant is nonzero, hence B is nondegenerate. $\square$

A second auxiliary local identity of the spin-factor construction is the Cayley–Hamilton identity

$$x \circ x - \operatorname{tr}_{\mathcal{S}}(x)x + N(x)\mathbf{1}_{\mathcal{S}} = 0, \quad \operatorname{tr}_{\mathcal{S}}(a, u) = 2a. \quad (2.1)$$

The cone of squares

$$C_{\mathcal{S}} := \{x \in \mathcal{S} : \exists y \in \mathcal{S}, x = y \circ y\} \quad (2.2)$$

has the explicit coordinate description

$$C_{\mathcal{S}} = \{(a, u) \in \mathcal{S} : a \geq 0, N(a, u) \geq 0\} = \{(a, u) \in \mathcal{S} : a \geq \|u\|\}. \quad (2.2a)$$

Thus $C_{\mathcal{S}}$ is the closed future Lorentz cone determined by the scalar coordinate on $\mathcal{S}$.

Definition 2.4 (Proper orthochronous Lorentz group). Let $G_{\text{Lor}} \subset O(B)$ be the subgroup of linear automorphisms $F : \mathcal{S} \rightarrow \mathcal{S}$ satisfying

$$N(Fx) = N(x), \quad x \in C_{\mathcal{S}} \iff Fx \in C_{\mathcal{S}}, \quad \det F = 1. \quad (2.3)$$

The definition depends only on $(\mathcal{S}, N, C_{\mathcal{S}})$. Relative to the future cone $C_{\mathcal{S}}$, it is the proper orthochronous Lorentz group of B . In particular, no matrix realization over $\mathbb{C}$, no pre-existing Lorentzian manifold, and no frame bundle enters the construction.

Remark 2.5 (Scope: local algebra only). At this point $\mathcal{S}$ is a four-dimensional real vector space with a Lorentz form. It is not a tangent space and there is no manifold, chart, atlas, base point, or covering space in the construction.

3 Clifford algebra, a norm-one group, and a two-fold Lorentz covering

3.1 The Euclidean Clifford algebra

Let

$$\text{Cl}_3 := \text{Cl}(V, q_3)$$

be the real Clifford algebra characterized by the relation

$$\iota(v)^2 = q_3(v)\mathbf{1}. \quad (3.1)$$

For the standard Euclidean basis $\varepsilon_0, \varepsilon_1, \varepsilon_2$ of V , write $\gamma_i := \iota(\varepsilon_i)$. Then

$$\gamma_i^2 = 1, \quad \gamma_i \gamma_j = -\gamma_j \gamma_i \quad (i \neq j). \quad (3.2)$$

The signs are those of the *positive* Euclidean quadratic form on V ; the Lorentzian signature has already appeared in the distinct four-dimensional spin factor $\mathcal{S}$.

Definition 3.1 (Paravector embedding). Define

$$A : \mathcal{S} \longrightarrow \text{Cl}_3, \quad A(a, v) := a\mathbf{1} + \iota(v). \quad (3.3)$$

Let c denote the Clifford conjugation used in the construction. Its restriction to paravectors changes the sign of the vector part. The key identity is

$$A(x)c(A(x)) = N(x)\mathbf{1}. \quad (3.4)$$

Proposition 3.2 (Jordan–Clifford compatibility). *For all $x, y \in \mathcal{S}$,*

$$A(x \circ y) = \frac{1}{2}(A(x)A(y) + A(y)A(x)). \quad (3.5)$$

Moreover, the image $A(\mathcal{S})$ generates Cl_3 as a real algebra.

Proof. Expand the product using

$$\iota(u)\iota(v) + \iota(v)\iota(u) = 2\langle u, v \rangle \mathbf{1}.$$

The scalar part becomes $2(ab + \langle u, v \rangle)$, while the vector part becomes $2(av + bu)$, proving (3.5). Since $A(0, v) = \iota(v)$, the image contains every Clifford generator; the universal generation property of the Clifford algebra then gives the second assertion. $\square$

The Clifford algebra Cl_3 has its standard universal property: any unital associative real algebra receiving a linear map $f : V \rightarrow \mathcal{A}$ with $f(v)^2 = q_3(v)\mathbf{1}$ receives a unique unital algebra homomorphism from Cl_3 ; see [1, 6].

3.2 The Clifford norm-one group

Definition 3.3 (Clifford norm-one group). An element $g \in \text{Cl}_3$ satisfies the Clifford norm-one condition when

$$g c(g) = c(g) g = 1. \quad (3.6)$$

The corresponding units form a subgroup

$$G_{\text{Spin}} \subset \text{Cl}_3^\times. \quad (3.7)$$

For $g \in G_{\text{Spin}}$, the Clifford action on paravectors preserves $A(\mathcal{S})$. Transferring this action through the embedding A defines a linear automorphism

$$\rho(g) : \mathcal{S} \longrightarrow \mathcal{S}.$$

For every $g \in G_{\text{Spin}}$, the automorphism $\rho(g)$ belongs to G_{Lor} , and

$$\rho : G_{\text{Spin}} \longrightarrow G_{\text{Lor}} \quad (3.8)$$

is a group homomorphism. After the covering theorem below, this is the paravector realization of the Lorentz double cover used throughout the paper.

Theorem 3.4 (Two-fold central covering). *The homomorphism ρ is surjective and*

$$\ker \rho = \{1, -1\}. \quad (3.9)$$

The kernel is central and has two elements. With the topologies inherited from the finite-dimensional Clifford model and the induced topology on G_{Lor} , both groups are Hausdorff topological groups, ρ is continuous, the kernel is discrete, and every $k \in G_{\text{Lor}}$ has an open neighbourhood U admitting a continuous local section

$$s : U \longrightarrow G_{\text{Spin}}, \quad \rho \circ s = \text{id}_U. \quad (3.10)$$

Proof. We separate surjectivity, the kernel calculation, and the local-section statement.

Surjectivity. Let $F \in G_{\text{Lor}}$ and put $p = F(1_{\mathcal{S}})$. Because F preserves both N and the cone of squares,

$$N(p) = 1, \quad p \in C_{\mathcal{S}}, \quad p_0 \geq 0.$$

Write $p = (a, u)$. From $a^2 - \langle u, u \rangle = 1$ and $a \geq 0$ one has $a \geq 1$. Hence the following square root in the cone of squares

$$r := \left(\sqrt{\frac{a+1}{2}}, \frac{u}{2\sqrt{(a+1)/2}} \right)$$

is well defined and satisfies $r \circ r = p$ and $N(r) = 1$. Set $\beta = A(r) \in \text{Cl}_3$. The Clifford norm identity makes β an element of G_{Spin} , and the Jordan–Clifford compatibility gives

$$\rho(\beta)(1_{\mathcal{S}}) = p.$$

Consequently

$$G := \rho(\beta)^{-1} F$$

fixes $1_{\mathcal{S}}$. The stabilizer of $1_{\mathcal{S}}$ acts on the spatial summand as an element $R \in SO(3)$. By Cartan–Dieudonné, R is a product of hyperplane reflections in unit spatial vectors. For a unit vector u , the Clifford generator $\iota(u)$ acts on the spatial summand as minus the corresponding hyperplane reflection. Since $\det R = 1$, an even number of reflections occurs; the accumulated minus sign therefore disappears, and the product of the corresponding Clifford generators is an element $\gamma \in G_{\text{Spin}}$ with $\rho(\gamma) = G$. Thus $\rho(\beta\gamma) = F$.

Kernel. Suppose $g \in G_{\text{Spin}}$ acts trivially on every paravector. Evaluating at $1_{\mathcal{S}}$ first shows that Clifford reversion and Clifford conjugation agree on g . The trivial-action equation then implies

$$gA(x) = A(x)g \quad \text{for every } x \in \mathcal{S}.$$

Because the paravector image generates Cl_3 as a real algebra, g commutes with all of Cl_3 . The centre computed in the Clifford model is $\mathbb{R} \oplus \mathbb{R}\omega$, where $\omega = \gamma_0\gamma_1\gamma_2$ and $\omega^2 = -1$. Hence $g = a + b\omega$. For central elements Clifford conjugation fixes g , so the norm-one condition gives

$$(a + b\omega)^2 = 1 \implies a^2 - b^2 = 1, \quad 2ab = 0.$$

The first equation excludes $a = 0$; therefore $b = 0$ and $a = \pm 1$. Conversely, ± 1 act trivially. Thus $\ker \rho = \{\pm 1\}$; centrality and cardinality two follow immediately.

Continuous local sections. This construction uses only the Clifford model. Define the linear map

$$\mathcal{F}(y) := y + \sum_{i=0}^2 \gamma_i y \gamma_i.$$

In the eight-monomial Clifford frame this kills the six noncentral monomials and multiplies the two central ones by 4, so $\mathcal{F}(y) \in \mathbb{R} \oplus \mathbb{R}\omega$. Next define the auxiliary map

$$\mathcal{R}(F) := A(F1_S) + \sum_{i=0}^2 A(F(0, \varepsilon_i)) \gamma_i.$$

For $g \in G_{\text{Spin}}$, direct Clifford multiplication gives

$$\mathcal{R}(\rho(g)) = g \mathcal{F}(\text{rev } g),$$

so $\mathcal{R}$ recovers g up to a central factor. Identify the centre $\mathbb{R} \oplus \mathbb{R}\omega$ with $\mathbb{C}$ by the real-algebra isomorphism

$$j : \mathbb{C} \xrightarrow{\sim} \mathbb{R} \oplus \mathbb{R}\omega, \quad j(i) = \omega.$$

Choose a real-linear map $\pi_{\mathbb{C}} : \text{Cl}_3 \rightarrow \mathbb{C}$ whose restriction to the centre is j^{-1} . Define the continuous auxiliary scalar function

$$\Delta(F) := \pi_{\mathbb{C}}(\mathcal{R}(F) c(\mathcal{R}(F))).$$

The key identity is

$$\mathcal{R}(\rho(g)) = j(w)g \implies \Delta(\rho(g)) = w^2.$$

Thus the auxiliary scalar function depends on the central factor through its square. At the identity, $\Delta(1) = 16$. On the open neighbourhood where $\Delta(F)$ lies in a slit plane avoiding the branch cut, the principal complex square root is continuous. Dividing the Clifford element $\mathcal{R}(F)$ by this square root leaves exactly the two choices differing by $\{\pm 1\}$, and yields a continuous G_{Spin} -valued section s with $\rho(s(F)) = F$ and $s(1) = 1$. Finally, for arbitrary $k \in G_{\text{Lor}}$, choose one lift u_k of k and left-translate the identity neighbourhood and its section. This produces an open neighbourhood of k with a continuous local section. A different choice of u_k changes that section only by the central kernel element -1 . $\square$

Thus one obtains the exact central extension

$$1 \longrightarrow \{\pm 1\} \longrightarrow G_{\text{Spin}} \xrightarrow{\rho} G_{\text{Lor}} \longrightarrow 1. \quad (3.11)$$

The symbol G_{Spin} denotes the Clifford norm-one group of Definition 3.3. Its relation to G_{Lor} is exactly the central extension (3.11); no separate identification with the even-subalgebra group $\text{Spin}(3) \subset \text{Cl}_3^0$ is used.

A classical complex-matrix realization provides a comparison, but it is not an input to theorem 3.4. In particular, the construction does not define G_{Spin} by first declaring it to be $\text{SL}(2, \mathbb{C})$.

4 The orthogonal Lie algebra and bivectors

The local group construction has a parallel infinitesimal object that can be defined without first making G_{Spin} into a Lie group.

Definition 4.1 (Orthogonal Lie algebra). Let

$$\mathfrak{so}(B) := \{T \in \text{End}_{\mathbb{R}}(\mathcal{S}) : B(Tx, y) + B(x, Ty) = 0 \text{ for all } x, y \in \mathcal{S}\}. \quad (4.1)$$

For $u, v \in \mathcal{S}$, define

$$K_{u,v}(x) := B(v, x)u - B(u, x)v. \quad (4.2)$$

Then $K_{u,v} \in \mathfrak{so}(B)$, the map is alternating in u, v , and it therefore induces a linear map

$$\Phi : \Lambda^2 \mathcal{S} \longrightarrow \mathfrak{so}(B), \quad u \wedge v \longmapsto K_{u,v}. \quad (4.3)$$

Theorem 4.2 (Canonical identification $\Lambda^2 \mathcal{S} \simeq \mathfrak{so}(B)$). *The map Φ is a linear isomorphism. Consequently*

$$\dim_{\mathbb{R}} \mathfrak{so}(B) = \dim_{\mathbb{R}} \Lambda^2 \mathcal{S} = 6. \quad (4.4)$$

Moreover, $\mathfrak{so}(B)$ is closed under the commutator

$$[A, C] := AC - CA, \quad (4.5)$$

so it is a six-dimensional real Lie algebra.

Proof. Skewness of $K_{u,v}$ follows by expanding (4.2) and using the symmetry of B . Closure of $\mathfrak{so}(B)$ under the commutator follows by applying the skewness identity twice:

$$B(ACx, y) = -B(Cx, Ay) = B(x, CAy),$$

and similarly with A and C interchanged.

It remains to prove that Φ is bijective. Let

$$\tau = (1, 0), \quad b_i = (0, \varepsilon_i), \quad i = 0, 1, 2,$$

where $\varepsilon_0, \varepsilon_1, \varepsilon_2$ is the fixed Euclidean basis of V . For $A \in \mathfrak{so}(B)$, introduce the six coefficients

$$\begin{aligned} c_0 &= (A\tau)_V^0, & c_1 &= (A\tau)_V^1, & c_2 &= (A\tau)_V^2, \\ c_3 &= \langle (Ab_0)_V, \varepsilon_1 \rangle, & c_4 &= \langle (Ab_0)_V, \varepsilon_2 \rangle, & c_5 &= \langle (Ab_1)_V, \varepsilon_2 \rangle. \end{aligned} \quad (4.6a)$$

The B -skew condition determines every other matrix coefficient from these six: it forces $(A\tau)_0 = 0$, determines the scalar component of $A(0, u)$ by $\langle (A\tau)_V, u \rangle$, and makes the spatial block antisymmetric. Hence if all six numbers in (4.6a) vanish, then $A\tau = 0$ and $Ab_i = 0$ for $i = 0, 1, 2$. Since τ, b_0, b_1, b_2 span $\mathcal{S}$, one has $A = 0$. Thus the six coefficients determine A uniquely.

Conversely, for $c = (c_0, \dots, c_5) \in \mathbb{R}^6$, define the explicit bivector

$$\begin{aligned} z(c) &:= -c_0 \tau \wedge b_0 - c_1 \tau \wedge b_1 - c_2 \tau \wedge b_2 \\ &\quad + c_3 b_0 \wedge b_1 + c_4 b_0 \wedge b_2 + c_5 b_1 \wedge b_2. \end{aligned} \quad (4.6b)$$

A direct evaluation of $K_{u,v}$ on the four basis vectors shows that the six coordinates of $\Phi(z(c))$ are exactly $c_0, \dots, c_5$. For any $A \in \mathfrak{so}(B)$, taking c to be the six coordinates of A therefore gives an endomorphism $\Phi(z(c))$ with the same six coordinates as A ; uniqueness of the six coefficients yields $\Phi(z(c)) = A$. Thus Φ is surjective.

For injectivity, the map $z : \mathbb{R}^6 \rightarrow \Lambda^2 \mathcal{S}$ in (4.6b) is injective because the six coordinates of $\Phi(z(c))$ recover c . Since

$$\dim \mathbb{R}^6 = 6 = \dim \Lambda^2 \mathcal{S},$$

this injective linear map is also surjective. If $\Phi(z) = \Phi(z')$, write $z = z(c)$ and $z' = z(c')$; applying the six coordinate functionals gives $c = c'$, hence $z = z'$. Therefore Φ is injective as well, and so it is a linear isomorphism. The dimension statement follows. $\square$

Remark 4.3 (Open problem: relation with the Lie algebra of G_{Spin}). The statement

$$\text{Lie}(G_{\text{Spin}}) \simeq \mathfrak{so}(B) \tag{4.7}$$

is not part of the present construction. The group G_{Spin} is established here as a topological group, not as a smooth Lie group. Accordingly, the differential

$$d\rho_e : \text{Lie}(G_{\text{Spin}}) \longrightarrow \mathfrak{so}(B) \tag{4.8}$$

has not been constructed. These statements remain open in the present paper.

5 From local model to global quotient space

The next step is deliberately not phrased as a consequence of the Spin group. The reason is structural: a fibre transition tells us how to compare fibres over a point already known to lie in two patches. It does not say which points of two local copies represent the same point of a global quotient space.

5.1 Gluing data

Let I be an index set. A local piece is an open subset $D_i \subset \mathcal{S}$.

Definition 5.1 (Gluing datum). A gluing datum B consists of:

1. open sets $D_i \subset \mathcal{S}$;
2. open overlap domains $W_{ij} \subset D_i$;
3. continuous maps $\phi_{ij} : \mathcal{S} \rightarrow \mathcal{S}$, used on W_{ij} ;

such that, on the indicated domains,

$$\begin{aligned} W_{ii} &= D_i, & \phi_{ii}(x) &= x, \\ \phi_{ij}(W_{ij}) &\subset W_{ji}, & \phi_{ji}(\phi_{ij}(x)) &= x, \end{aligned} \tag{5.1}$$

and whenever $x \in W_{ij}$ and $\phi_{ij}(x) \in W_{jk}$,

$$x \in W_{ik}, \quad \phi_{jk}(\phi_{ij}(x)) = \phi_{ik}(x). \tag{5.2}$$

This is an explicit *sufficient* gluing datum. We do not claim that every field in Definition 5.1 is separately necessary, or that this is the unique minimal encoding of overlap information.

Form the tagged disjoint union

$$X_0 := \coprod_{i \in I} D_i \tag{5.3}$$

and define

$$(i, x) \sim (j, y) \iff x \in W_{ij} \text{ and } \phi_{ij}(x) = y. \tag{5.4}$$

The axioms above make $\sim$ an equivalence relation. Define

$$M_B := X_0 / \sim \tag{5.5}$$

with the quotient topology.

Proposition 5.2 (Topological quotient construction). *The quotient map $X_0 \rightarrow M_B$ is open. Each local piece D_i embeds homeomorphically onto an open subset of M_B , and these images cover M_B . Thus the images of the D_i form a topological atlas of open coordinate domains modelled on $\mathcal{S}$, before any smoothness assertion is made.*

Proof. The saturation of an open subset of X_0 is open because each overlap set W_{ij} is open and ϕ_{ij} is a homeomorphism between W_{ij} and W_{ji} , with inverse ϕ_{ji} . Hence the quotient map is open. Equality of two points from the same tagged piece forces equality of their local coordinates by $\phi_{ii} = \text{id}$, so the map from each D_i is injective. Openness and continuity then give an open embedding. Surjectivity of the quotient map proves that the chart images cover. $\square$

Hausdorffness is not a formal consequence of Definition 5.1; the usual line-with-two-origins phenomenon already shows why a separation condition is needed.

Definition 5.3 (Closed gluing relation). The equivalence relation associated with B has closed graph when

$$\Gamma_B := \{(p, q) \in X_0 \times X_0 : p \sim q\} \quad (5.6)$$

is closed.

Proposition 5.4 (Hausdorffness and second countability). *If Γ_B is closed, then M_B is Hausdorff. If, in addition, I is countable, then M_B is second countable.*

Proof. Let $q : X_0 \rightarrow M_B$ be the quotient map. By Proposition 5.2, q is continuous, open, and surjective. Consequently

$$q \times q : X_0 \times X_0 \longrightarrow M_B \times M_B$$

is again an open quotient map. The preimage of the diagonal is exactly the gluing relation:

$$(q \times q)^{-1}(\Delta_{M_B}) = \{(p, q) : p \sim q\} = \Gamma_B.$$

If Γ_B is closed, the quotient property therefore implies that Δ_{M_B} is closed. This is equivalent to Hausdorffness.

For second countability, each D_i is an open subspace of the finite-dimensional real vector space $\mathcal{S}$, hence is second countable. If I is countable, the tagged disjoint union $X_0 = \coprod_i D_i$ is second countable. An open quotient of a second-countable space is second countable: the images under q of a countable basis form a countable basis of M_B . Hence M_B is second countable. $\square$

5.2 Why the quotient space requires additional data

The need for gluing data is itself a theorem, not merely a warning.

Theorem 5.5 (Local pieces do not determine the quotient space). *There exist two gluing data over the same two indexed local domains such that the local pieces agree exactly but their quotient spaces are not homeomorphic.*

Proof. Take a nonempty connected open local domain $D \subset \mathcal{S}$ and two copies indexed by $\{\text{false}, \text{true}\}$. In the first gluing, set $W_{ij} = D$ for every pair of indices and use the identity identification. In the second, take $W_{ij} = \emptyset$ for $i \neq j$ and again use the identity where defined. The first quotient is connected; the second is the disjoint union of two nonempty open components and is not connected. Hence no homeomorphism can exist between them. $\square$

Corollary 5.6. *No datum that only prescribes a fibre transition law over already specified overlaps can, by itself, determine the missing identifications between distinct local pieces.*

5.3 Smooth gluing

For the constructed atlas, the coordinate change from piece i to piece j is precisely ϕ_{ij} on W_{ij} .

Definition 5.7 (Smooth gluing). A gluing datum is smoothly compatible when

$$\phi_{ij}|_{W_{ij}}$$

is C^∞ for every i, j .

Theorem 5.8 (Conditional smooth four-manifold). *Suppose:*

1. B is a gluing datum over $\mathcal{S}$;
2. its gluing relation is closed;

3. the index set I is countable;

4. B satisfies Definition 5.7.

Then M_B is Hausdorff, second countable, and a C^∞ four-manifold modelled on $\mathcal{S}$.

Proof. The Hausdorff and second-countable assertions are exactly Proposition 5.4. It remains to identify the smooth structure of the actual quotient atlas.

For each nonempty piece D_i , the open embedding of Proposition 5.2 yields a chart whose target is D_i . On an overlap of the i - and j -charts, the source of the coordinate change is not merely contained in the specified overlap set: it is exactly W_{ij} . If $y \in W_{ij}$, the quotient relation identifies (i, y) with $(j, \phi_{ij}(y))$, and therefore the coordinate change is literally

$$y \longmapsto \phi_{ij}(y).$$

Thus every transition map of the constructed atlas is a restriction of one of the functions already required to be C^∞ by Definition 5.7. The transition maps are therefore smooth, so the resulting atlas is a smooth atlas and M_B is a smooth manifold. Finally, the model space is $\mathcal{S} = \mathbb{R} \oplus \mathbb{R}^3$, hence $\dim_{\mathbb{R}} \mathcal{S} = 4$.

Notice that the argument proves sufficiency of the smooth-gluing hypothesis for this constructed atlas. It does not claim the converse or necessity of every specified smoothness condition. $\square$

Proposition 5.9 (Counterexample: topological gluing need not be smooth). *There are gluing data satisfying the topological gluing axioms whose transition is a homeomorphism but not differentiable at the origin. An explicit example is*

$$H(t, x)_0 = t, \quad H(t, x)_k = x_k + |t| \quad (k = 1, 2, 3),$$

with inverse

$$H^{-1}(t, x)_0 = t, \quad H^{-1}(t, x)_k = x_k - |t|.$$

Both maps are continuous, so the topological quotient construction applies. But along the line $t \mapsto (t, 0)$, every shifted spatial component contains $|t|$, which is not differentiable at $t = 0$. Hence the smooth-gluing condition fails.

Remark 5.10 (Sufficiency, not necessity). For the quotient construction used here, smooth gluing is proved sufficient. A converse is not asserted. In particular, quotient representative selection may cause the constructed atlas to omit some index-labelled chart even when the corresponding specified ϕ_{ij} is non-smooth.

6 Čech 1-cocycles for the Clifford norm-one group and tangent transitions

Fix from now on a smooth gluing B satisfying the hypotheses needed to obtain $M = M_B$.

6.1 Tangent transitions

Because the actual atlas transition is ϕ_{ij} , the tangent bundle transition is its derivative.

Proposition 6.1 (Tangent transition law). *For $y \in W_{ij}$, the tangent coordinate change of the constructed manifold is*

$$T_{ij}(y) := D\phi_{ij}(y) : \mathcal{S} \longrightarrow \mathcal{S}. \quad (6.1)$$

It is a linear isomorphism, with inverse

$$D\phi_{ji}(\phi_{ij}(y)). \quad (6.2)$$

The transitions satisfy the derivative cocycle law by the chain rule.

The invertibility in (6.2) is derived from $\phi_{ji} \circ \phi_{ij} = \text{id}$ on the open overlap domain rather than inserted as a separate hypothesis.

Remark 6.2 (Local coordinate identifications do not identify the cocycles). Although each chart identifies tangent vectors locally with the model space $\mathcal{S}$, these local identifications do not identify the tangent transition cocycle with the independent Lorentz cocycle. The relevant transition law is (6.1).

6.2 A group-valued Čech 1-cocycle

Let $\mathcal{U} = \{U_i\}_{i \in I}$ be the cover of M by the chart images. A G_{Spin} -valued Čech 1-cocycle consists of continuous maps

$$\tilde{g}_{ij} : U_i \cap U_j \longrightarrow G_{\text{Spin}} \quad (6.3)$$

satisfying the Čech cocycle law

$$\tilde{g}_{ij}(x)\tilde{g}_{jk}(x) = \tilde{g}_{ik}(x) \quad (6.4)$$

on triple overlaps. Projection gives the Lorentz-valued cocycle

$$g_{ij} := \rho \circ \tilde{g}_{ij} : U_i \cap U_j \longrightarrow G_{\text{Lor}}. \quad (6.5)$$

This is a fibre transition cocycle on the same cover, but it is not the tangent transition cocycle by definition.

Theorem 6.3 (The tangent and Lorentz transition cocycles need not coincide). *There exist a smooth two-chart gluing B and a G_{Spin} -valued Čech 1-cocycle $\tilde{g}$ such that*

$$g_{01}(x) = \text{id}_{\mathcal{S}} \quad \text{for every } x, \quad (6.6)$$

while

$$T_{01}(y) = 2 \text{id}_{\mathcal{S}} \quad \text{for every } y. \quad (6.7)$$

Hence the gluing data and the G_{Spin} -valued cocycle do not force $T_{ij} = g_{ij}$.

Proof. Take two copies of the whole local model with

$$\phi_{01}(y) = 2y, \quad \phi_{10}(y) = \tfrac{1}{2}y.$$

All overlap domains are the whole model, and the gluing is smooth. Thus $D\phi_{01} = 2 \text{id}$. Independently, choose the pointwise trivial G_{Spin} -valued Čech 1-cocycle $\tilde{g}_{ij} = 1$; its projection is the identity. Evaluating at $\mathbf{1}_{\mathcal{S}}$ distinguishes the two transformations. $\square$

Remark 6.4. The conclusion is one of non-derivability, not non-existence. The same rescaling example admits compatible soldering data. Equality of transition matrices is therefore not the correct invariant criterion for coupling the two bundles.

7 A soldering isomorphism between the Lorentz vector bundle and the tangent bundle

Let $E_g \rightarrow M$ denote the rank-four real vector bundle defined by the Lorentz-valued Čech 1-cocycle g_{ij} . The additional object considered in this section is a fibrewise linear identification $s : E_g \rightarrow TM$ whose local representatives vary smoothly and intertwine the two transition cocycles. The inverse map $\theta := s^{-1} : TM \rightarrow E_g$ has the conventional direction of an E_g -valued solder form [5, 9].

7.1 Pointwise fibre isomorphisms

At the weakest level one may choose, for each chart and each point of its image in M , a linear isomorphism

$$e_i(x) : \mathcal{S} \xrightarrow{\sim} T_x M \quad (7.1)$$

such that on $U_i \cap U_j$,

$$e_j(x)v = e_i(x)(g_{ij}(x)v). \quad (7.2)$$

This algebraic condition is sufficient to transport the Lorentz form B to each tangent fibre, but it imposes no regularity on the field $x \mapsto e_i(x)$.

Proposition 7.1 (Counterexample: pointwise fibre isomorphisms need not be continuous). *There exist pointwise families of fibre isomorphisms obeying (7.2) whose chart representative is discontinuous. Consequently pointwise fibre isomorphisms alone do not produce a continuous vector-bundle isomorphism or a smooth Lorentzian metric.*

Explicit construction. Take the one-chart quotient with identity gluing and the trivial G_{Spin} -valued Čech 1-cocycle. In this case (7.2) imposes no nontrivial overlap constraint. Let x_0 be the chart point represented by the zero vector and define the frame field

$$e(x) = \begin{cases} 2 \text{ id}_{\mathcal{S}}, & x = x_0, \\ \text{id}_{\mathcal{S}}, & x \neq x_0. \end{cases} \quad (7.2a)$$

Every fibre map is a linear isomorphism, so this is a valid pointwise family of fibre isomorphisms. Its local representative, however, jumps from id to 2id at x_0 , and is therefore discontinuous. The discontinuity is also visible in the transported quadratic form: evaluated on the distinguished unit, the coefficient is 1 away from x_0 , whereas at x_0 the inverse frame contributes a factor $1/2$, giving $1/4$. Thus even the induced fibrewise metric coefficient is discontinuous. This discontinuous pointwise frame field is therefore not induced by soldering data of Definition 7.2. $\square$

7.2 Local representatives of a soldering isomorphism

We describe the soldering isomorphism by local representatives in the chart coordinates. Let

$$A_i(y) \in \text{GL}(\mathcal{S})$$

be a continuous linear automorphism for $y \in D_i$.

Definition 7.2 (Local soldering data). Local soldering data for $(B, \tilde{g})$ are a family

$$A_i : D_i \longrightarrow \text{GL}(\mathcal{S}) \quad (7.3)$$

such that:

1. the map $y \mapsto A_i(y)$, viewed in the finite-dimensional space of continuous linear maps, is C^∞ on D_i ;
2. on $y \in W_{ij}$,

$$A_j(\phi_{ij}(y)) = T_{ij}(y) \circ A_i(y) \circ g_{ij}(x_y), \quad (7.4)$$

where $x_y \in M$ is the quotient point represented by (i, y) .

Equation (7.4) is the local intertwining identity. The tangent map T_{ij} converts i -chart tangent components to j -chart tangent components, while the Čech convention for E_g identifies components in the two local trivializations by g_{ij} . The equation is therefore the compatibility condition between the two transition cocycles.

The inverse family $A_i(y)^{-1}$ is automatically smooth; no separate smoothness assumption for the inverse is needed.

Theorem 7.3 (Soldering isomorphism). *The local soldering data determine a vector-bundle isomorphism*

$$s : E_g \longrightarrow TM$$

covering id_M , at the following regularity level:

1. continuous total-space map and continuous inverse;
2. linear isomorphism on each fibre;
3. C^∞ local representatives A_i ;
4. C^∞ local representatives A_i^{-1} for the inverse;
5. exact intertwining of the Lorentz-valued and tangent transition cocycles.

Conversely, a vector-bundle isomorphism with these properties yields local soldering data of Definition 7.2.

Proof. Write E_g in the local trivializations determined by the cocycle g_{ij} , and write TM in the tangent trivializations of the constructed atlas. On the i -th chart define the local total-space map

$$(y, v) \longmapsto (y, A_i(y)v). \quad (7.4a)$$

If $y \in W_{ij}$, changing the source trivialization by g_{ij} and the target trivialization by T_{ij} gives, by (7.4), the same point as first applying (7.4a) in chart i . Hence the local maps agree on equivalent representatives and descend to a globally defined map of total spaces covering id_M . Each fibre map is a linear isomorphism because each $A_i(y)$ is.

The local representative of the descended map in the i -th source and target trivializations is exactly A_i . Since A_i is smooth, it is continuous, and the standard local-trivialization criterion gives continuity of the total-space map. Pointwise inversion on $\text{GL}(\mathcal{S})$ is smooth; hence $y \mapsto A_i(y)^{-1}$ is smooth. The same argument applied to these inverse representatives yields a continuous inverse total-space map. This proves the five stated properties.

Conversely, suppose such a vector-bundle isomorphism is given. Read its local representative in the chart trivializations and call it $A_i(y)$. Fibrewise linear isomorphism gives $A_i(y) \in \text{GL}(\mathcal{S})$, smoothness of the local representatives gives C^∞ -regularity, and equality of the globally defined map when read in the two overlapping trivializations is precisely the intertwining identity (7.4). Thus the local representatives form local soldering data of the stated kind. The two constructions are inverse at the level asserted in the theorem. $\square$

Remark 7.4 (Exact regularity level). We establish a vector-bundle isomorphism with continuous total-space maps in both directions, fibrewise linear isomorphisms, and smooth local representatives for the map and its inverse. No smooth structure on the total spaces is used to

state a stronger smooth bundle isomorphism. In particular, the limitation concerns the level at which the global equivalence is stated, not the smoothness of the projected Lorentz transition representatives, which is forced in chart coordinates by the soldering data.

Indeed, (7.4) can be solved for g_{ij} :

$$g_{ij}(x_y) = A_i(y)^{-1} \circ T_{ij}(y)^{-1} \circ A_j(\phi_{ij}(y)), \quad (7.5)$$

up to the convention used in (7.4). Hence the soldering data force the projected Lorentz transition to possess smooth local representatives in chart coordinates even though the underlying G_{Spin} -valued Čech 1-cocycle is only assumed continuous.

7.3 Gauge freedom of the soldering isomorphism

For fixed B and fixed G_{Spin} -valued Čech 1-cocycle, the soldering data need not be unique. Let $\text{Gau}(TM)$ denote the group of smooth vector-bundle automorphisms of TM covering id_M . It acts on the set of local soldering data.

Theorem 7.5 (Torsor of soldering isomorphisms). *Whenever the set of soldering isomorphisms represented by local data for fixed $(B, \tilde{g})$ is nonempty, the action of $\text{Gau}(TM)$ on it is free and transitive.*

Proof. Let $A = \{A_i\}$ and $A' = \{A'_i\}$ be two choices of local soldering data for the same gluing datum and the same G_{Spin} -valued Čech 1-cocycle. Define

$$a_i(y) := A'_i(y) A_i(y)^{-1}. \quad (7.6)$$

The field a_i is smooth because multiplication and inversion in the finite-dimensional general linear group are smooth. Applying the intertwining identity (7.4) once to A and once to A' , and cancelling the common Lorentz-valued transition, gives

$$a_j(\phi_{ij}(y)) T_{ij}(y) = T_{ij}(y) a_i(y), \quad (7.7)$$

which is exactly the gauge-compatibility law for TM . By construction,

$$(a \cdot A)_i(y) = a_i(y) A_i(y) = A'_i(y),$$

so the action is transitive.

For freeness, if $a \cdot A = a' \cdot A$, then $a_i(y) A_i(y) = a'_i(y) A_i(y)$ at every point. Right cancellation by the invertible map $A_i(y)$ gives $a_i(y) = a'_i(y)$ for all i, y , hence $a = a'$. Pointwise composition and inversion preserve (7.7), so these fields form a group and the action is free and transitive. $\square$

Thus existence is an additional requirement, while uniqueness is neither expected nor claimed.

8 The induced Lorentzian metric

Let $A = \{A_i\}$ be a choice of soldering data.

8.1 Metric coefficients

In chart i , define

$$g_i(y)(u, v) := B(A_i(y)^{-1}u, A_i(y)^{-1}v). \quad (8.1)$$

Theorem 8.1 (Induced Lorentzian metric). *The fields g_i satisfy:*

1. *for fixed $u, v \in \mathcal{S}$, the function $y \mapsto g_i(y)(u, v)$ is C^∞ on D_i ;*
2. *$g_i(y)$ is symmetric and nondegenerate;*
3. *on every overlap domain,*

$$g_j(\phi_{ij}(y))(T_{ij}(y)u, T_{ij}(y)v) = g_i(y)(u, v); \quad (8.2)$$

4. *in the $A_i(y)$ -frame, the form has signature $(1, 3)$.*

Hence the g_i are the local coefficient fields of a smooth section

$$g \in \Gamma(\text{Sym}^2 T^*M)$$

that is nondegenerate of signature $(1, 3)$ at every point. Thus g is a smooth Lorentzian metric on M in the standard semi-Riemannian sense [10, 9].

Proof. Smoothness follows from smoothness of A_i^{-1} and bilinearity of B . Symmetry and non-degeneracy are inherited from B through the linear isomorphism $A_i(y)$. For the transformation law, use (7.4) to express the inverse j -frame after application of T_{ij} in terms of the inverse i -frame and the projected Lorentz-valued transition. Since $g_{ij}(x) \in G_{\text{Lor}}$ preserves B , these Lorentz factors cancel and yield (8.2). Finally,

$$g_i(y)(A_i(y)a, A_i(y)b) = B(a, b),$$

so Proposition 2.3 transfers the signature. $\square$

The important logical point is that the metric is *derived after* soldering data are supplied. It is not a field of the gluing datum and is not inserted into the local input.

8.2 Orientation compatibility

Let

$$F_{ij}(y) := A_j(\phi_{ij}(y))^{-1} \circ T_{ij}(y) \circ A_i(y). \quad (8.3)$$

By the intertwining identity (7.4) this is the projected Lorentz transition (with the inverse-index convention induced by (7.4)). Therefore

$$\det F_{ij}(y) = 1. \quad (8.4)$$

Proposition 8.2 (Orientation compatibility induced by soldering data). *The local tangent frames obtained from the maps A_i have transition matrices of determinant 1. Equivalently, if $a_i = \det A_i$, then on every overlap*

$$\det T_{ij}(y) = \frac{a_j(\phi_{ij}(y))}{a_i(y)},$$

so the multiplicative Čech 1-cocycle formed by the determinants of the tangent transition maps is a continuous coboundary.

8.3 Compatible future cones

Define the interior future cone by

$$\mathcal{C}_{\text{int}}^+ := \{x \in \mathcal{S} : x \in C_{\mathcal{S}} \text{ and } N(x) > 0\}.$$

This uses only the already constructed cone of squares and quadratic form. In chart i , define

$$\mathcal{C}_i^+(y) := A_i(y)\mathcal{C}_{\text{int}}^+. \quad (8.5)$$

Proposition 8.3 (Future-cone compatibility). *For $y \in W_{ij}$,*

$$u \in \mathcal{C}_i^+(y) \iff T_{ij}(y)u \in \mathcal{C}_j^+(\phi_{ij}(y)). \quad (8.6)$$

Thus the soldering data determine, in each tangent fibre, a continuous chart-independent choice of one of the two timelike cone components, namely the component corresponding to $\mathcal{C}_{\text{int}}^+$.

The local vector field

$$\tau_i(y) := A_i(y)\mathbf{1}_{\mathcal{S}} \quad (8.7)$$

is smooth, future-directed, and unit timelike. It need not glue to a single global field. More precisely, the distinguished fields τ_i glue across an overlap exactly when the corresponding projected Lorentz transition fixes $\mathbf{1}_{\mathcal{S}}$.

Remark 8.4 (Scope: continuous choice of a timelike cone component). The conclusion established here is the continuous compatible choice, in each tangent fibre, of the timelike cone component corresponding to $\mathcal{C}_{\text{int}}^+$ (8.6). No theorem in this paper identifies this choice with every standard equivalent formulation of time orientability. Likewise, existence of an arbitrary global smooth future-directed timelike section would require an additional global section theorem for the bundle constructed here.

9 Čech lifting obstruction for the central extension and tangent–Lorentz compatibility

We now analyze the lifting problem for the Lorentz-valued cocycle on the chosen cover.

9.1 Lifting obstruction on the chosen cover

Let

$$g_{ij} : U_i \cap U_j \rightarrow G_{\text{Lor}}$$

be a Lorentz-valued Čech 1-cocycle. Assume continuous lifts

$$h_{ij} : U_i \cap U_j \rightarrow G_{\text{Spin}}, \quad \rho(h_{ij}) = g_{ij}. \quad (9.1)$$

have been chosen on the double overlaps. The local-section theorem for ρ guarantees such lifts only locally in the target; existence of a continuous lift on an entire double overlap is additional information, or may require a suitable refinement of the cover. On triple overlaps, define the defect

$$\epsilon_{ijk} := h_{ij} h_{jk} h_{ik}^{-1}. \quad (9.2)$$

Because $\rho(\epsilon_{ijk}) = 1$,

$$\epsilon_{ijk} \in \ker \rho = \{\pm 1\}. \quad (9.3)$$

This defines a $\{\pm 1\}$ -valued Čech 2-cocycle class on the chosen cover $\mathcal{U}$. The independence from the chosen lift family is elementary at this level. If h'_{ij} is another lift of the same Lorentz-valued cocycle, then there are $\{\pm 1\}$ -valued functions $\alpha_{ij} : U_i \cap U_j \rightarrow \{\pm 1\}$ with

$$h'_{ij} = \alpha_{ij} h_{ij}.$$

Because the kernel is central,

$$\begin{aligned} \epsilon'_{ijk} &= h'_{ij} h'_{jk} (h'_{ik})^{-1} \\ &= \alpha_{ij} \alpha_{jk} \alpha_{ik}^{-1} \epsilon_{ijk}. \end{aligned} \quad (9.3a)$$

Thus two representatives differ by the $\{\pm 1\}$ -valued Čech coboundary $\delta\alpha$, and determine the same Čech 2-cocycle class on the chosen cover. A G_{Spin} -valued Čech 1-cocycle satisfying the cocycle law gives $\epsilon_{ijk} = 1$ identically, so its class is trivial.

This is the familiar mechanism underlying the usual obstruction theory for Spin structures [11, 8, 6]; however, the class considered here is defined only on the chosen cover $\mathcal{U}$.

Remark 9.1 (Scope: no Stiefel–Whitney identification). No theorem here identifies the determinant condition derived from the transition maps on the chosen cover with $w_1(TM)$, or the $\{\pm 1\}$ -valued lifting class on the chosen cover with $w_2(TM)$. No refinement limit or comparison with global $H^1(M; \mathbb{Z}/2)$ or $H^2(M; \mathbb{Z}/2)$ is used. Consequently the statement $w_1(TM) = w_2(TM) = 0$ is not a theorem of the present paper.

9.2 Intertwining under soldering data

The intertwining identity (7.4) identifies the projected Lorentz-valued cocycle with the tangent transition cocycle expressed in the local frames defined by A_i .

Theorem 9.2 (Intertwining of the tangent and Lorentz cocycles). *For every choice of soldering data and every overlap, the change of tangent components between the local frames defined by A_i is exactly the Lorentz-valued transition g_{ij} .*

This is a rearrangement of the intertwining equation (7.4), but it is the precise intertwining relation between the two cocycles.

Theorem 9.3 (Consequences of the intertwining identity). *Assume B is smoothly glued and soldering data exist for a G_{Spin} -valued Čech 1-cocycle $\tilde{g}$. Then:*

1. *there are continuous nowhere-zero functions $a_i : D_i \rightarrow \mathbb{R}$ such that*

$$\det T_{ij}(y) a_i(y) = a_j(\phi_{ij}(y)); \quad (9.4)$$

2. *the tangent transition cocycle in the local frames defined by A_i equals $\rho(\tilde{g}_{ij})$;*
3. *every local lift family of that projected cocycle has trivial $\{\pm 1\}$ -valued Čech 2-cocycle class for the central extension on the chosen cover;*
4. *the compatible future timelike cone field of Proposition 8.3 exists.*

Proof. For (1), take the determinant of the intertwining identity (7.4). The Lorentz-valued factor has determinant one, so the determinant of the tangent transition is the ratio of the determinants of the local maps A_i , giving (9.4). Statement (2) is theorem 9.2. For (3), the G_{Spin} -valued Čech 1-cocycle itself is a coherent lift of its projection; the triple-overlap defect is therefore trivial. Independence of the obstruction from the chosen local lift family transfers triviality to every lift family. Statement (4) is Proposition 8.3. $\square$

Two useful contrapositives are immediate.

Corollary 9.4 (Necessary determinant condition for the existence of soldering data). *If no continuous nowhere-zero 0-cochain a_i can satisfy (9.4), then no soldering data exist for any G_{Spin} -valued Čech 1-cocycle over that gluing.*

Corollary 9.5 (Obstruction to lifting the Lorentz cocycle through the central extension). *If the $\{\pm 1\}$ -valued Čech 2-cocycle class associated with the central extension is nontrivial on the chosen cover, then it admits no coherent G_{Spin} -valued Čech 1-cocycle projecting to it. In particular, it cannot be the Lorentz-valued cocycle obtained from soldering data whose lift is a coherent G_{Spin} -valued Čech 1-cocycle.*

The converse of theorem 9.3 is not proved. Vanishing of these two necessary conditions is therefore not asserted to imply the existence of soldering data.

10 Examples, obstructions, and nonuniqueness of lifts

The following examples distinguish necessary conditions from features of specific constructions.

10.1 One-chart example

On a one-chart quotient there are no nontrivial overlap constraints. The gluing is smooth and soldering data exist. Thus neither of the preceding obstructions on the chosen cover occurs in this example.

10.2 Orientation-reversing counterexample

There is a smooth orientation-reversing two-chart gluing whose cross-overlap has two relevant components. On the wrap-around component the tangent transition has determinant -1 , whereas on the ordinary overlap component it has determinant $+1$.

Proposition 10.1 (Orientation reversal prevents the existence of soldering data). *For the orientation-reversing example, no soldering data exist for any G_{Spin} -valued Čech 1-cocycle.*

Reason. Every projected Lorentz-valued transition lies in G_{Lor} and hence has determinant $+1$. If soldering data existed, the determinant of each local map A_i would be a continuous nowhere-zero function on a connected chart domain and therefore would have constant sign there. The determinant of the intertwining identity (7.4) on the wrap-around component, where the tangent determinant is -1 , forces opposite signs in the two charts; the same law on the ordinary component, where the tangent determinant is $+1$, forces the same signs. These requirements are incompatible. This is the concrete mechanism behind Corollary 9.4; the presence of both overlap components is essential. $\square$

G_{Spin} -valued Čech 1-cocycles nevertheless exist over this same gluing. Thus existence of a G_{Spin} -valued Čech 1-cocycle is strictly weaker than existence of soldering data intertwining its Lorentz projection with TM .

10.3 Combinatorial labels do not determine the existence of a soldering isomorphism

Consider also a deliberately abstract combinatorial label structure. For arbitrary such combinatorial labels, the same label input is compatible with the one-chart example, where local soldering data exist, and with the orientation-reversing two-chart example, where no soldering data exist for any G_{Spin} -valued Čech 1-cocycle. Therefore these combinatorial labels do not determine the existence of soldering data.

10.4 Nonuniqueness of lifts modulo kernel-valued Čech 0-cochains

Because $\ker \rho = \{\pm 1\}$, distinct lifts of the same Lorentz-valued cocycle may remain inequivalent under $\{\pm 1\}$ -valued Čech 0-cochains on the chosen cover.

Proposition 10.2 (Two inequivalent lifts with identical Lorentz projection). *On the chosen one-loop cover there are two G_{Spin} -valued Čech 1-cocycles that:*

1. *have the same projected Lorentz-valued cocycle;*
2. *both admit soldering data;*
3. *are not equivalent under the equivalence relation generated by $\{\pm 1\}$ -valued Čech 0-cochains on the chosen cover.*

Proposition 10.3 (Independent sign choices on a finite loop family). *For N independent loop labels, the construction gives an injection of $\{\pm 1\}^N$ into the equivalence classes under $\{\pm 1\}$ -valued Čech 0-cochains on the chosen cover. In particular, for $N = 2$ there are four pairwise distinguished classes, all compatible with soldering data in the periodic gluing example.*

Proof of Propositions 10.2 and 10.3. For each loop n , the cross-overlap of its two patches has two connected components: the wrap-around component and the ordinary component. Given a sign assignment $\sigma \in \{\pm 1\}^N$, define a kernel-valued Čech 1-cocycle z^σ which equals σ_n on the wrap-around component of loop n and equals 1 on the ordinary component and on all unaffected overlaps. Since every value of z^σ lies in $\ker \rho = \{\pm 1\}$, twisting the trivial $G_{\text{Spin-}}$ -valued Čech 1-cocycle by z^σ leaves the projected Lorentz cocycle unchanged. Consequently the same soldering data for the periodic gluing example work for every sign assignment.

It remains to prove that distinct sign assignments are not related by a gauge transformation by $\{\pm 1\}$ -valued Čech 0-cochains on the chosen cover. Suppose $\sigma_n \neq \tau_n$ for some loop n , and assume that patchwise kernel-valued gauge functions λ_- and λ_+ relate the corresponding twisted cocycles. The kernel is discrete and each patch is connected, so continuity forces λ_- and λ_+ to be constant on their patches. Evaluating the gauge law at a point of the ordinary overlap, where both sign cocycles equal 1, gives

$$\lambda_- \lambda_+^{-1} = 1.$$

Evaluating the same law at a point of the wrap-around component gives, using centrality of the kernel,

$$\tau_n = \lambda_- \sigma_n \lambda_+^{-1} = \sigma_n,$$

contradicting $\sigma_n \neq \tau_n$. Hence different sign assignments give different equivalence classes under $\{\pm 1\}$ -valued Čech 0-cochains on the chosen cover. For one loop this yields exactly the two classes of Proposition 10.2; for N independent loops it gives the asserted injection, and for $N = 2$ the four sign assignments give four pairwise inequivalent classes. $\square$

Remark 10.4 (Scope). The preceding statements do not compute $H^1(M; \mathbb{Z}/2)$. They concern equivalence under $\{\pm 1\}$ -valued Čech 0-cochains on the chosen cover; no refinement argument identifying these equivalence classes with global cohomology is part of the present construction.

Hence the projected Lorentz-valued cocycle and soldering data do not determine a unique lift to G_{Spin} on the chosen cover.

11 A one-parameter family of Čech 1-cocycles

We next study an explicit one-parameter family of G_{Spin} -valued Čech 1-cocycles on a fixed periodic gluing. No connection one-form, path-ordered transport, holonomy functional, curvature two-form, or field equation is introduced.

11.1 A one-parameter subgroup

Choose the first Euclidean basis direction $\varepsilon_0 \in V$ and its Clifford image

$$\gamma_0 := \iota(\varepsilon_0), \quad \gamma_0^2 = 1.$$

The generator γ_0 is chosen for this example; no theorem makes this direction canonical.

Definition 11.1 (One-parameter subgroup of G_{Spin}). For $\lambda \in \mathbb{R}$, define

$$G(\lambda) := \cosh\left(\frac{\lambda}{2}\right) + \sinh\left(\frac{\lambda}{2}\right) \gamma_0 \in \text{Cl}_3. \quad (11.1)$$

The hyperbolic identities and $\gamma_0^2 = 1$ give

$$G(\lambda)c(G(\lambda)) = 1,$$

hence $G(\lambda) \in G_{\text{Spin}}$. Moreover,

$$G(0) = 1, \quad G(\lambda + \mu) = G(\lambda)G(\mu), \quad G(-\lambda) = G(\lambda)^{-1}. \quad (11.2)$$

The parameterization is injective.

The projection $\rho(G(\lambda))$ is nontrivial for $\lambda \neq 0$. On the plane spanned by the scalar direction and $(0, \varepsilon_0)$, its action is the standard boost block

$$\begin{pmatrix} \cosh \lambda & \sinh \lambda \\ \sinh \lambda & \cosh \lambda \end{pmatrix}, \quad (11.3)$$

and it acts trivially on the two orthogonal Euclidean directions. In particular,

$$\rho(G(\lambda))\mathbf{1}_S = (\cosh \lambda, \sinh \lambda \varepsilon_0). \quad (11.4)$$

Remark 11.2 (Scope of the example). The family (11.1) is a deliberately selected one-parameter subgroup generated by one chosen Clifford direction. It is neither proved unique nor canonical, and it does not exhaust other one-parameter subgroups or cocycle families.

11.2 The periodic gluing example

Fix the periodic gluing, together with the chosen cover and two-component periodic overlap. We make no identification of this quotient with $S^1 \times \mathbb{R}^3$, and no fundamental-group theorem is required.

The G_{Spin} -valued Čech 1-cocycle is varied using $G(\lambda)$. The gluing and hence the tangent transitions are held fixed; the projected Lorentz-valued transition is obtained by applying ρ to the G_{Spin} -valued transition. We then ask for local soldering data satisfying the intertwining identity (7.4).

Let

$$\Lambda_{\text{sold}} := \{\lambda \in \mathbb{R} : \text{soldering data exist for the cocycle at } \lambda\}. \quad (11.5)$$

Theorem 11.3 (Existence of soldering data for every parameter). *For the fixed periodic gluing,*

$$\Lambda_{\text{sold}} = \mathbb{R}. \quad (11.6)$$

Thus every finite parameter value admits explicitly constructed soldering data. In particular existence is symmetric under $\lambda \mapsto -\lambda$.

Proof. Let $\chi : \mathbb{R} \rightarrow [0, 1]$ be a fixed C^∞ transition profile with $\chi(t) = 0$ for $t \leq 3$ and $\chi(t) = 1$ for $t \geq 4$. On the two patches of each periodic loop define

$$\alpha_\lambda(p, y) = \begin{cases} 0, & p \text{ is the first patch,} \\ \lambda \chi(y_0), & p \text{ is the second patch.} \end{cases} \quad (11.6a)$$

Let $P(t) = \rho(G(t)) \in G_{\text{Lor}}$. By the one-parameter subgroup law, $P(s+t) = P(s)P(t)$ and $P(-t) = P(t)^{-1}$, and the closed form (11.3) makes $t \mapsto P(t)$ smooth. Define

$$A_p^{(\lambda)}(y) := P(\alpha_\lambda(p, y)). \quad (11.6b)$$

This is the identity on the first patch and smoothly interpolates on the second patch between the identity and $P(\lambda)$.

All tangent transitions of the fixed periodic gluing are the identity. The projected Lorentz-valued transition on an overlap is $P(\beta_{pq}^{(\lambda)}(x))$, where $\beta_{pq}^{(\lambda)}$ is $\pm\lambda$ on the wrap-around component and 0 on the ordinary component. By the defining values of χ , one verifies on each overlap component that

$$\alpha_\lambda(q, \phi_{pq}(y)) = \alpha_\lambda(p, y) + \beta_{pq}^{(\lambda)}(x_y). \quad (11.6c)$$

Therefore

$$\begin{aligned} A_q^{(\lambda)}(\phi_{pq}(y)) &= P(\alpha_\lambda(q, \phi_{pq}(y))) \\ &= P(\alpha_\lambda(p, y))P(\beta_{pq}^{(\lambda)}(x_y)) \\ &= A_p^{(\lambda)}(y) g_{pq}^{(\lambda)}(x_y), \end{aligned}$$

which is precisely the intertwining identity (7.4) because the tangent transition is id. Smoothness and pointwise invertibility follow from those of P and χ . Hence soldering data exist for every $\lambda \in \mathbb{R}$, proving $\Lambda_{\text{sold}} = \mathbb{R}$. $\square$

11.3 Gauge equivalence by group-valued Čech 0-cochains

Let $\tilde{g}^{(\lambda)}$ denote the G_{Spin} -valued Čech 1-cocycle at parameter λ .

Theorem 11.4 (Gauge equivalence of the one-parameter family). *For every pair $\lambda_1, \lambda_2 \in \mathbb{R}$ there are continuous maps $u_i : U_i \rightarrow G_{\text{Spin}}$ such that, on each overlap,*

$$\tilde{g}_{ij}^{(\lambda_2)} = u_i \tilde{g}_{ij}^{(\lambda_1)} u_j^{-1}. \quad (11.7)$$

The G_{Spin} -valued Čech 0-cochain u is continuous. After projection by ρ , the resulting Lorentz-valued Čech 0-cochain is C^∞ on every chart domain, and the projected Lorentz cocycles are gauge equivalent by that projected 0-cochain.

Proof. First compare an arbitrary λ with the identity member $\lambda = 0$. Using the same interpolation function α_λ as in (11.6a), define on patch p

$$u_p^{(\lambda)}(x) := G(-\alpha_\lambda(p, \text{coord}_p(x))). \quad (11.7a)$$

The map is continuous because $G(t)$ is continuous in the topology on G_{Spin} and the chart-coordinate profile is continuous. On an overlap, (11.6c) and the group law of G give

$$\tilde{g}_{pq}^{(\lambda)}(x) = u_p^{(\lambda)}(x) \tilde{g}_{pq}^{(0)}(x) (u_q^{(\lambda)}(x))^{-1}. \quad (11.7b)$$

Thus every parameter value is gauge equivalent under G_{Spin} -valued Čech 0-cochains to the identity cocycle, i.e. the member with $\lambda = 0$. For two arbitrary values λ_1, λ_2 , compose the inverse gauge from λ_1 to the identity member with the gauge from the identity member to λ_2 .

Applying ρ to (11.7b) gives the corresponding Lorentz gauge relation. Its chartwise representative is $P(-\alpha_\lambda)$; by the explicit hyperbolic formula (11.3) and smoothness of α_λ , this projected gauge is C^∞ on every chart domain. No smooth Lie-group structure on G_{Spin} is used or claimed. $\square$

The equivalence relation generated only by $\{\pm 1\}$ -valued Čech 0-cochains on the chosen cover behaves differently. For $\lambda \neq 0$, the cocycle at parameter λ is not obtained from the identity cocycle merely by a $\{\pm 1\}$ -valued 0-cochain, because its Lorentz projection is nontrivial. There is no contradiction: the two equivalence relations answer different questions.

11.4 Bijection between sets of local soldering data

For a parameter λ , write

$$\mathcal{E}_\lambda := \left\{ \text{soldering data for } (B_{\text{per}}, \tilde{g}^{(\lambda)}) \right\}. \quad (11.8)$$

Theorem 11.5 (Bijection between sets of local soldering data). *For every $\lambda_1, \lambda_2 \in \mathbb{R}$, there is an explicit bijection*

$$\mathcal{T}_{\lambda_1, \lambda_2} : \mathcal{E}_{\lambda_1} \xrightarrow{\sim} \mathcal{E}_{\lambda_2}, \quad (11.9)$$

obtained by precomposing each local representative A_i with the projected gauge relating the two G_{Spin} -valued Čech 1-cocycles. The inverse is $\mathcal{T}_{\lambda_2, \lambda_1}$.

Proof. Set

$$\delta_{12}(p, y) := \alpha_{\lambda_2}(p, y) - \alpha_{\lambda_1}(p, y), \quad Q_{12,p}(y) := P(\delta_{12}(p, y)). \quad (11.9a)$$

For soldering data $A \in \mathcal{E}_{\lambda_1}$, define

$$(\mathcal{T}_{\lambda_1, \lambda_2} A)_p(y) := A_p(y) \circ Q_{12,p}(y). \quad (11.9b)$$

The fields $Q_{12,p}$ are smooth and invertible. Subtracting the two identities (11.6c) for λ_1 and λ_2 gives exactly the overlap equation needed to convert the intertwining identity at λ_1 into the intertwining identity at λ_2 . Thus (11.9b) is soldering data at λ_2 .

Since

$$\delta_{21} = -\delta_{12} \quad \text{and} \quad P(-t) = P(t)^{-1},$$

transporting from λ_1 to λ_2 and then back composes with $Q_{12}Q_{21} = \text{id}$ pointwise. Hence $\mathcal{T}_{\lambda_2, \lambda_1}$ is the inverse of $\mathcal{T}_{\lambda_1, \lambda_2}$, proving the stated bijection. $\square$

Theorem 11.6 (Metric preservation). *Let $A \in \mathcal{E}_{\lambda_1}$, and let $A' = \mathcal{T}_{\lambda_1, \lambda_2}(A)$. Then for every $x \in M$ and $X, Y \in T_x M$,*

$$g_{A'}(X, Y) = g_A(X, Y). \quad (11.10)$$

Proof. In a chart, (11.9b) gives

$$A'_p = A_p \circ Q_{12,p}, \quad (A'_p)^{-1} = Q_{12,p}^{-1} \circ A_p^{-1}.$$

Therefore

$$\begin{aligned} g_{A',p}(u, v) &= B\left(Q_{12,p}^{-1}A_p^{-1}u, Q_{12,p}^{-1}A_p^{-1}v\right) \\ &= B\left(A_p^{-1}u, A_p^{-1}v\right) \\ &= g_{A,p}(u, v), \end{aligned}$$

because $Q_{12,p}(y) = P(\delta_{12}(p, y)) \in G_{\text{Lor}}$ preserves B . The equality holds in every chart and is compatible with the tangent transition law, hence it is the equality of the induced Lorentzian metrics at every base point and on every pair of tangent vectors. $\square$

Corollary 11.7 (Gauge equivalence and equality of the induced Lorentzian metrics). *All members of the one-parameter family are gauge equivalent under G_{Spin} -valued Čech 0-cochains. Under the explicit bijections between the sets of local soldering data, corresponding choices of soldering data induce the same Lorentzian metric.*

Remark 11.8 (Scope of Corollary 11.7). The result does not classify arbitrary deformations of G_{Spin} -valued Čech 1-cocycles, arbitrary covers, co-varying bases, or connection-level deformations. It does not imply gauge equivalence for arbitrary cocycle families, and it carries no curvature interpretation. The parameter λ is the coordinate of the selected one-parameter subgroup of G_{Spin} (11.1), not a curvature scalar.

12 Scope and open problems

The results above determine the following scope and open problems.

From the local input $V = \mathbb{R}^3$ with its Euclidean form, one has constructed the real spin factor with Lorentzian quadratic form, the Euclidean Clifford algebra, the two-fold central covering $G_{\text{Spin}} \rightarrow G_{\text{Lor}}$, and the six-dimensional orthogonal Lie algebra $\mathfrak{so}(B)$. After adding explicit gluing data and sufficient topological and smooth conditions, one obtains a smooth four-manifold. After adding soldering data, one obtains a smooth Lorentzian metric, orientation compatibility, a continuous compatible choice of the timelike cone component corresponding to C_{int}^+ in each tangent fibre, and the exact intertwining of the tangent transition cocycle with the projected Lorentz-valued cocycle.

The following structures and identifications have not been constructed in this paper.

1. The topological group G_{Spin} has not been given a smooth Lie-group structure and related to $\mathfrak{so}(B)$ as a Lie group/Lie algebra pair:

$$\text{Lie}(G_{\text{Spin}}) \simeq \mathfrak{so}(B) \quad \text{is open.} \quad (12.1)$$

2. The differential $d\rho_e$ of the covering homomorphism has not been constructed.
3. No connection data associated with either the G_{Spin} -valued or the Lorentz-valued Čech cocycle have been introduced.
4. Consequently there is no covariant derivative, parallel transport, holonomy, curvature, torsion, action functional, or field equation in the present construction.
5. The determinant Čech 1-cocycle and the $\{\pm 1\}$ -valued Čech 2-cocycle class on the chosen cover have not been identified with $w_1(TM)$ and $w_2(TM)$, respectively.
6. The compatible future timelike cone field constructed in Proposition 8.3 has not been identified by theorem with all standard formulations of time orientability.

In the periodic gluing example of section 11, the entire one-parameter family is gauge equivalent under G_{Spin} -valued Čech 0-cochains, and the induced Lorentzian metric is unchanged under the explicit bijection between the sets of local soldering data. Thus this particular transition-cocycle parameter does not define distinct induced Lorentzian metrics at the level studied here. No conclusion about connection-level geometry follows from this example.

13 Conclusion

The results establish a separation between levels of geometric information.

At the local level, from the specified positive Euclidean three-space one constructs the real spin factor $\mathcal{S}$ with its Lorentzian quadratic form, the Euclidean Clifford algebra, the two-fold central covering $G_{\text{Spin}} \rightarrow G_{\text{Lor}}$, and the orthogonal Lie algebra $\mathfrak{so}(B)$. This part is determined by the local input.

At the global level, gluing information is independent. Local pieces, even when equipped with fibre transitions, do not determine which points of different pieces represent the same point of the quotient. Explicit gluing data are therefore additional input. When the graph of the gluing equivalence relation is closed, the index set is countable, and the transition maps are smooth, the quotient is a smooth four-manifold.

At the tangent-bundle level, a second independent datum is required. The derivative cocycle $D\phi_{ij}$ of that manifold is not forced to equal the independently specified Lorentz-valued cocycle. Local soldering data provide the vector-bundle isomorphism $s : E_g \rightarrow TM$ intertwining the associated Lorentz vector bundle with TM . These data yield a smooth Lorentzian metric, orientation compatibility, a continuous compatible choice of the timelike cone component corresponding to C_{int}^+ in each tangent fibre, and triviality, on the chosen cover, of the $\{\pm 1\}$ -valued Čech 2-cocycle class associated with the central extension.

The resulting implication is therefore conditional and exact:

$$\begin{aligned}
 (\mathbb{R}^3, \langle \cdot, \cdot \rangle) &\implies (\mathcal{S}, N, \text{Cl}_3, G_{\text{Spin}} \rightarrow G_{\text{Lor}}, \mathfrak{so}(B)) \\
 &\quad \text{local algebraic data + gluing} \\
 &\quad + \text{closed gluing relation + countable index + smoothness} \implies M^4 \\
 &\quad M^4 + G_{\text{Spin}}\text{-valued Čech 1-cocycle + soldering data} \\
 &\quad \Downarrow \\
 &\quad (TM, g), \\
 &\quad \text{orientation compatibility,} \\
 &\quad \text{compatible choice of a timelike cone component in each tangent fibre,} \\
 &\quad \text{tangent cocycle in the local frames defined by } A_i = \text{projected Lorentz cocycle,} \\
 &\quad \text{trivial } \{\pm 1\}\text{-valued Čech 2-cocycle class for the central extension on the chosen cover.}
 \end{aligned} \tag{13.1}$$

Each implication in (13.1) is conditional on the displayed hypotheses.

In the periodic gluing example of section 11, the selected members of the one-parameter family are gauge equivalent for all finite parameters. The sets of local soldering data at different parameter values are explicitly bijective, and corresponding choices of soldering data induce the same Lorentzian metric. This result is restricted to the specified family and does not address connection-level geometry.

A Dependency and status table

Object or statement	Status	Dependency
$V = \mathbb{R}^3$, Euclidean form	input	none
$\mathcal{S}, N, B$, signature (1, 3)	DERIVED	Euclidean input
Cl_3 , paravectors	DERIVED	q_3
$G_{\text{Spin}} \rightarrow G_{\text{Lor}}$, kernel $\{\pm 1\}$	DERIVED	Clifford algebra and paravector action
$\mathfrak{so}(B) \simeq \Lambda^2 \mathcal{S}$	DERIVED	B
Gluing data	ADDITIONAL DATUM	not determined by local pieces
Closed graph of the gluing equivalence relation	additional hypothesis	used in the Hausdorff proof
Countable index set	additional hypothesis	used in the second-countability proof
Smooth gluing	sufficient additional hypothesis	base transitions ϕ_{ij}
Smooth M^4	DERIVED	preceding global hypotheses
Tangent transitions $D\phi_{ij}$	DERIVED	smooth atlas
G_{Spin} -valued Čech 1-cocycle	global datum	chosen cover of M
Soldering data	ADDITIONAL DATUM	existence not automatic
Smooth Lorentzian metric	DERIVED	soldering data
Orientation compatibility	DERIVED	soldering data
Compatible choice of a timelike cone component in each tangent fibre	DERIVED	soldering data
$\{\pm 1\}$ -valued Čech 2-cocycle class trivial on the chosen cover	DERIVED	soldering data + G_{Spin} -valued Čech 1-cocycle
w_1, w_2 comparison	OPEN	characteristic-class comparison theorem
$\text{Lie}(G_{\text{Spin}}) \simeq \mathfrak{so}(B)$	OPEN	smooth Lie-group structure on G_{Spin}
$d\rho_e$	OPEN	preceding infinitesimal bridge
Connection, holonomy, curvature	OPEN	additional connection data

B Source correspondence

This appendix gives the correspondence between the mathematical statements in the paper and the Lean source. Implementation identifiers retain the historical project naming; these names are not mathematical terminology used in the paper.

Paper statement	Principal Lean declaration(s)
Spin factor and Lorentz form	<code>SpinCore.sJ_jordan</code> , <code>SpinCore.lorentz_quadratic_form</code>
Jordan–Clifford bridge	<code>SpinCore.clifford_jordan</code> , <code>SpinCore.adjoin_spinToCl_eq_top</code>
Two-fold central covering	<code>SpinCore.spin_double_cover_bundled</code> , <code>SpinCore.spin_double_cover_topological_endpoint</code> , <code>SpinCore.spin_double_cover_local_section_endpoint</code>
Bivectors and $\mathfrak{so}(B)$	<code>SpinCore.bivectorEquivSkew</code> , <code>SpinCore.finrank_gB</code> , <code>SpinCore.gBLie_bracket</code>
Local pieces do not determine the quotient space	<code>EmergentBase.base_not_determined_by_local_pieces</code>
Topological quotient construction	<code>EmergentBase.BaseGluingData.emergentManifoldCertificate</code>
Smooth manifold	<code>EmergentBase.BaseGluingData.isManifold_of_smoothGluing</code> , <code>EmergentBase.BaseGluingData.smoothEmergentManifoldCertificate</code>
Tangent transition	<code>EmergentBase.BaseGluingData.tangentTransition_eq_derivative_baseTransition</code>
Independence of tangent and Lorentz-valued cocycles	<code>SpinNative.tangent_spin_base_data_do_not_force_transition_identification</code>
Soldering data and vector-bundle isomorphism	<code>SpinNative.SmoothTangentSolderData</code> ; <code>SpinNative.smooth_solder_iff_regular_bundle_equivalence</code>
Smooth Lorentzian metric	<code>SpinNative.SmoothTangentSolderData.smooth_tangent_lorentz_metric</code>
Orientation compatibility and compatible timelike cone choice	<code>SpinNative.SmoothTangentSolderData.orientation_time_orientation_status</code>
Intertwining of tangent and Lorentz-valued cocycles	<code>Task36.classical_reconvergence</code> , <code>Task36.spin_obstruction_nonzero_implies_no_solder</code>
Counterexamples to existence of soldering data	<code>Task36.adversarial_global_topology_certificate</code> , <code>Task36.spin_foam_labels_do_not_determine_solder</code>
One-parameter G_{Spin} -valued Čech 1-cocycle family	<code>Task37.Deformation.transportState_add</code> , <code>Task37.Deformation.projectedTransportState_apply</code>
Existence of soldering data for the periodic family	<code>Task37.Deformation.loopParaSolder (construction)</code> , <code>Task37.Deformation.controlAAdmissible_all</code> , <code>Task37.Deformation.regularRegion_loopTransportFamily</code>
Gauge equivalence under G_{Spin} -valued Čech 0-cochains	<code>Task37.Deformation.loopGaugeFun (construction)</code> , <code>Task37.Deformation.spinCocycle_gaugeEquiv</code> , <code>Task37.Deformation.projectedCocycle_gaugeEquiv</code>
Combined theorem for the one-parameter family	<code>Task37.Deformation.native_transport_knob_smoketest_certificate</code>
Bijection between sets of local soldering data	<code>Task37.Deformation.loopSolderEquiv (construction)</code> , <code>Task37.Deformation.loopSolderTransport_involutive</code> , <code>Task37.Deformation.loopSolderSolutionSpace_correspondence</code> , <code>Task37.Deformation.tangentMetric_loopSolderTransport</code>

The formal development is available at https://github.com/antaris82/The_Relational_Spin-Closure_Hypothesis [14]. The main development is written in Lean 4 and uses Mathlib [12, 13]. The mathematical claims of this paper should be read at the strength of the displayed hypotheses above; the Lean source supplies machine-checked proofs of the corresponding statements.

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AI and formalization disclosure

ChatGPT (OpenAI), acting as AI Assistant, was used for mathematical exploration, source-level analysis, organization, consistency checking, and editorial drafting. Aristotle was used as Lean Agent for prover-assisted formalization and verification. The editor curated the scope and the presentation. Mathematical claims are asserted only at the strength stated in the paper and are tied, where applicable, to the corresponding Lean declarations. The Lean source contains the machine-checked proofs of those statements.

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